Cruise Terminals. Ventilation design must ensure that fumes and odors from forklifts and cargo in work areas do not penetrate occupied and administrative areas.

Bus Terminals. The primary concerns with enclosed bus loading areas are health and safety problems, which must be handled by proper ventilation (see Chapter 16). Although diesel engine fumes are generally not as noxious as gasoline fumes, bus terminals often have many buses loading and unloading at the same time, and the total amount of fumes and odors may be disturbing.

In terms of health and safety, enclosed bus loading areas and automobile parking garages present the most serious problems. Three major problems are encountered, the first and most serious of which is emission of carbon monoxide (CO) by cars and oxides of nitrogen (NO_x) by buses, which can cause serious illness and possibly death. Oil and gasoline fumes, which may cause nausea and headaches and can create a fire hazard, are also of concern. The third issue is lack of air movement and the resulting stale atmosphere caused by increased CO content in the air. This condition may cause headaches or grogginess. Most codes require a minimum ventilation rate to ensure that the CO concentration does not exceed safe limits. Chapter 16 covers ventilation requirements and calculation procedures for enclosed vehicular facilities in detail.

All underground garages should have facilities for testing the CO concentration or should have the garage checked periodically. Problems such as clogged duct systems; improperly operating fans, motors, or dampers; or clogged air intake or exhaust louvers may not allow proper air circulation. Proper maintenance is required to minimize any operational defects.

3. WAREHOUSES AND DISTRIBUTION CENTERS

General Design Considerations

Warehouses can be defined as facilities that provide proper environment for the purpose of storing goods and materials. They are also used to store equipment and material inventory at industrial facilities. At times, warehouses may be open to the public. The buildings are generally not air conditioned, but often have sufficient heat and ventilation to provide a tolerable working environment. In many cases, associated facilities occupied by office workers, such as shipping, receiving, and inventory control offices, are air conditioned. Warehouses must be designed to accommodate the loads of materials to be stored, associated handling equipment, receiving and shipping operations and associated trucking, and needs of operating personnel. Types of warehouses include the following:

- **Heated and unheated general warehouses** provide space for bulk, rack, and bin storage, aisle space, receiving and shipping space, packing and crating space, and office and toilet space. As indicated some areas are typically equipped with small-decentralized air-conditioning systems for the support personnel.
- **Conditioned general warehouses** are similar to heated and unheated general warehouses, but can provide space cooling to meet the stored goods' requirements.
- **Refrigerated warehouses** are designed to preserve the quality of perishable goods and general supply materials that require refrigeration. This includes freeze and chill spaces, processing facilities, and mechanical areas. For information on this type of warehouse, see Chapters 23 and 24 in the 2018 *ASHRAE Handbook—Refrigeration*.
- **Controlled humidity (CH) and dry-air storage warehouses** are similar to general warehouses except that they are constructed with vapor barriers and contain humidity control equipment to maintain humidity at desired levels. For additional information, see Chapter 29 of Harriman et al. (2001).
- **Specialty warehouses** includes storing facilities with special and in some instances strict requirements for temperature, humidity, cleanliness, minimum ventilation rates, etc. These facilities are typically conditioned to achieve the required space conditions. These warehouses can be found in industrial and manufacturing facilities or can be standalone buildings. Examples include
  - Pharmaceutical and life sciences facilities. Good manufacturing practices (GMP) may be required.
  - Liquid storage (fuel and nonpropellants), flammable and combustible storage, radioactive material storage, hazardous chemical storage, and ammunition storage.
  - Automated storage and retrieval systems (AS/RS), which are designed for maximum storage and minimum personnel on site. They are built for lower-temperature operation with minimal heat and light needed, but require a tall structure with extremely level floors. In some cases, specialty HVAC equipment is required for servers and other computer areas in AS/RS facility.

Features already now common in warehouse designs are higher bays, sophisticated materials-handling equipment, broadband connectivity access, and more distribution networks. A wide range of storage alternatives, picking alternatives, material-handling equipment, and software exist to meet the physical and operational requirements. Warehouse spaces must also be flexible to accommodate future operations and storage needs as well as mission changes.

Areas that can be found in warehouses and distribution centers include the following:

- Storage areas
- Office and administrative areas
- Loading docks
- Light industrial spaces
- Computer/server rooms

Other areas can be site specific.

Design Criteria

Design criteria (temperature, humidity, noise, etc.) for warehouses are space specific; the designer should refer to the relevant sections and chapters (e.g., the section on Office Buildings for office and administration areas). For conditioned storage areas, the special requirements of the product stores dictate the design conditions.

Outdoor air for ventilation of office, administration, and support areas should be based on local code requirements or ASHRAE *Standard* 62.1. For general warehouses where special ventilation or minimum ventilation rates are not specifically defined, *Standard* 62.1 can be used as the criterion for minimum outdoor air. To define the specific ventilation and exhaust design criteria, consult local applicable ventilation and exhaust standards. Table 6-1 of *Standard* 62.1 recommends 0.3 $L/(s \cdot m^2)$ of ventilation as a design criterion for warehouse ventilation, although this amount may be insufficient when stored materials have harmful emissions.

Load Characteristics

Given the variety of warehouses facilities, every case should be analyzed carefully. In general, internal loads from lighting, people, and miscellaneous sources are low. Most of the load is thermal transmission and infiltration. An air-conditioning load profile tends to flatten where materials stored are massive enough to cause the peak load to lag. In humid climates, special attention should be given to the sensible and latent loads' variations for cases where the warehouse or distribution center is conditioned or cooled by thermostatically controlled packaged HVAC equipment. In these climates, it is common to satisfy the space temperature (i.e., very low or no sensible cooling load), but, because of infiltration of moist air and without proper cooling (i.e., the cooling equipment is off), for space humidity to be unacceptably high.

Variable Volume (constant face velocity). Hood has an opening or bypass designed to provide a prescribed minimum air intake when sash is closed, and an exhaust system designed to vary airflow in accordance with sash opening. Sash may be vertical, horizontal, or a combination of both.

Application: Moderate to highly hazardous processes; varying procedures.

Auxiliary Air (approximately constant-volume airflow). A plenum above the face receives air from a secondary air supply that provides partially conditioned or unconditioned outdoor air.

Note: Many organizations restrict the use of this type of hood.

Low Velocity or Reduced Flow (approximately constant-volume airflow with variable face velocity or variable volume). These hoods are designed to provide containment at lower average face velocities.

Application: Moderate to highly hazardous processes; varying procedures.

Filtered, Recirculating (approximately constant-volume airflow). Particulate filtration combined with chemical adsorption to remove contaminants.

Application: Moderate process with predictable procedures.

Radioisotope. Hood with special integral work surface, linings impermeable to radioactive materials, and structure strong enough to support high-density shielding materials. The interior must be constructed to prevent radioactive material buildup and allow complete cleaning. Ductwork should have flanged gasketed joints with quick-disconnect fasteners that can be readily dismantled for decontamination. High-efficiency particulate air (HEPA) and/or charcoal filters may be needed in exhaust duct.

Application: Laboratories using radioactive isotopes.

Perchloric Acid. Hood with special integral work surfaces, coved corners, and nonorganic lining materials. Perchloric acid is an extremely active oxidizing agent. Its vapors can form unstable deposits in the ductwork that present a potential explosion hazard. To alleviate this hazard, the exhaust system must be equipped with an internal water washdown and drainage system, and the ductwork must be constructed of smooth, impervious, cleanable materials that are resistant to acid attack. The internal washdown system must completely flush the ductwork, exhaust fan, discharge stack, and fume hood inner surfaces. Ductwork should be kept as short as possible with minimum elbows. Perchloric acid exhaust systems with longer duct runs may need a zoned washdown system to avoid water flow rates in excess of the capacity to drain water from the hood. Because perchloric acid is an extremely active oxidizing agent, organic materials should not be used in the exhaust system in places such as joints and gaskets. Ducts should be constructed of a stainless steel material, with a chromium and nickel content not less than that of 316L stainless steel, or of a suitable nonmetallic material. Joints should be welded and ground smooth. A perchloric acid exhaust system should only be used for work involving perchloric acid.

Application: Process and research laboratories using perchloric acid. Mandatory use because of explosion hazard.

California. Special hood with sash openings on multiple sides (usually horizontal).

Application: For enclosing large and complex research apparatus that require access from two or more sides.

Floor-Mounted Hood (Walk-In). Hood with sash openings to the floor. Sash can be either horizontal or vertical.

Application: For enclosing large or complex research apparatus. Not designed for personnel to enter while operations are in progress.

Distillation. Fume hood with extra depth and 1/3- to 1/2-height benches.

Application: For enclosing tall distillation apparatus.

Process (approximately constant-volume airflow with approximately constant face velocity). Standard hood with a fixed opening and without a sash. Not a fume hood. Considered a ventilated enclosure.

Application: Low-hazard processes; known procedures.

Canopy. Open hood with an overhead capture structure.

Application: Not a fume hood. Useful for heat or water vapor removal from some work areas. Not to be substituted for a fume hood. Not recommended when workers must bend over the source of heat or water vapor.

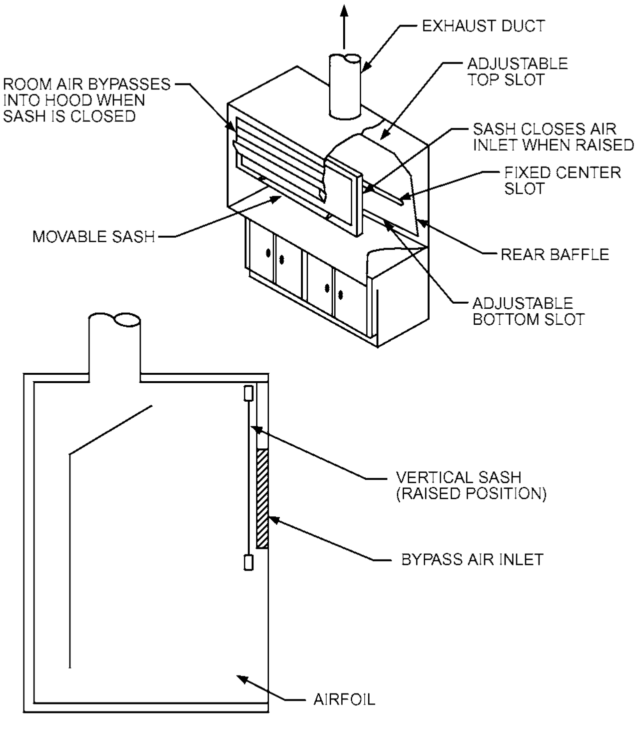

Fig. 1 Bypass Fume Hood with Vertical Sash and Bypass Air Inlet

Fume Hood Sash Configurations

The work opening has operable glass sash(es) for observation and shielding. A sash may be vertically operable, horizontally operable, or a combination of both. A vertically operable sash can incorporate single or multiple vertical panels. A horizontally operable sash incorporates multiple panels that slide in multiple tracks, allowing the open area to be positioned across the face of the hood. The combination of a horizontally operable sash mounted within a single vertically operable sash section allows the entire hood face to be opened for setup. The opening area can then be limited by closing the vertical panel, with only the horizontally sliding sash sections used during experimentation. Both the multiple vertical sash section and combination sash arrangement allow the use of larger fume hoods with limited opening areas, resulting in reduced exhaust airflow requirements. Fume hoods with vertically rising sash sections should include provisions around the sash to prevent the bypass of ceiling plenum air into the fume hood.

The design engineer should be involved from the start of project planning to ensure that space layout does not present unnecessary problems. For both functional organization and design efficiency, spaces in similar environmental zones should be kept physically contiguous, with the occupied collections zone and the unoccupied storage environments adjacent to one another. This allows not only for efficient ducting, but also for efficient interior wall (thermal and vapor barrier) construction. Appropriate zoning should minimize the necessity for downstream subzone equipment (VAV/reheat, subzone humidification) in most occupied and unoccupied collections zones.

Activities that pose a potential threat to collections environments (food areas, loading docks, fabrication and rough shops, maintenance and housekeeping, etc.) should be kept away from collections zones, with exhaust from these areas located away and downwind from any fresh air intakes for collections systems and the rest of the building. One key exception to this practice may be relative-humidity-controlled crate storage, which may share a zone with occupied collections areas. Loading docks should ideally be positively pressurized, and shared walls with collections zones should be carefully designed to mitigate any potential energy transfer, whether from a load perspective or to reduce risk of condensation or other issues. Where possible, a separate loading dock may be specified for cleaner transfer of collections materials and crates, with direct access to crate storage and collections intake work areas.

System Design and Envelope Performance

System designs that call for interior moisture control, whether humidification or dehumidification, must carefully consider the likely performance of the existing or designed envelope for both thermal insulation and vapor transfer. These issues are commonly discussed for exterior envelopes, but the same issues can exist with excessive thermal or vapor differentials across interior walls.

Thermal insulation/barriers should be adequate to moderate heat transfer through the structure, with particular attention paid to thermal bridges (e.g., wall studs, floor/wall junctions, roof/wall junctions) that may be sources of heat gain/loss or potential condensation points.

Vapor incursion or loss occurs by two primary paths: air leakage/gaps and diffusion. Air leakage or gaps speed heat transfer, but may be defeated for thermal loads with positive pressurization. Vapor carried by air cannot be fully mitigated pressurization: it slows the process by slowing air transfer, but vapor flow and equilibration between interior and exterior spaces continue independently based on vapor, rather than air, pressure. The differential between interior and exterior vapor pressures also drives diffusion (movement of water vapor through permeable materials) from areas of higher vapor pressure to lower. Diffusion manifests differently depending on exterior climates and interior environments. When exterior moisture conditions are higher than interior conditions, whether typically or seasonally, vapor may move inward, especially during dehumidification, adding to system load and causing efflorescence or other damage on interior surfaces. During humidification, vapor is typically forced outward through the envelope, and may limit the possible control of the interior environment and damage the structure's exterior. Diffusion in either direction, combined with interior wall temperatures, can cause significant structural damage ranging from mold growth in warm environments to structural failure in situations of repetitive freeze/thaw cycles.

Windows, doors, and skylights can pose risks as potential condensation points, as well as points of heat transfer. Repetitive condensation on doors and windows places that element at risk, and the potential transfer of that moisture to the wall or floor can lead to issues ranging from mold and wood rot to cracking and masonry damage because of freeze/thaw.

Older structures, especially those with historic significance, can be particularly problematic, especially in projects that propose indoor moisture control. Where envelope performance is in question, the first step is to reassess environmental goals: considering either higher or lower relative humidity ranges to minimize differences in vapor pressure or, in cold climates, lowering temperature to raise relative humidity rather than using mechanical humidification. If specific conditions are necessary for long-term preservation, alternatives for containment of preservation environments should be considered, such as

- Microenvironments (cabinetry, certain housing solutions, small-scale freezing)
- Offsite, purpose-built storage, which may provide better environmental control with better energy efficiency
- Limited envelope upgrades in part of the structure, often called **box-in-box** solutions, where interior surfaces are built in to provide room for installation of thermal and vapor barriers

Reliability and Resiliency

Designs should include some way to manage the interior environment during periods of interrupted operation, such as maintenance, equipment failure, power outages, and disasters.

Redundancy and back-up equipment may be applicable in the largest of cultural heritage applications, but capital cost, load cycling for long-term operability, and limited availability of physical space are all factors against redundancy as a typical practice. Rather, discussion commonly turns to reliability and resiliency as risk mitigation strategies for interrupted operation. Better understanding of collection equilibration times to environmental changes and improved building and envelope construction often mean that collections, particularly storage areas, may be able to hold appropriate environmental conditions for longer periods than previously understood. The design team should discuss scenarios, with designs reflecting potential response to events to minimize downtime. Spare equipment may be kept on site to allow for timely repairs, modulating outdoor air dampers may be set to fail closed to minimize infiltration, and in certain environments architectural and systems allowances may be made to provide natural ventilation in long-term disaster or recovery scenarios.

Stand-alone generators may be part of this strategy, but their design and siting should be reviewed carefully for load capacity, accessibility, weight restrictions (for rooftop applications), fuel availability, and locations of fuel storage tanks, which pose their own significant disaster risk.

Loads

Accounting for design capacities for both sensible and latent energy is critical in cultural heritage applications and collections environments. Though common when specifying some systems types such as four-pipe subcool/reheat and desiccant designs, adequate latent capacity is often disregarded with two-pipe and standard package unit designs, resulting in environments that may be able to maintain temperature control, but commonly see relative humidity conditions rise beyond safe limits as exterior temperatures climb and cooling capacity is dedicated to the sensible load.

Certain specific load characteristics of cultural heritage buildings should be considered in system design. Occupancy patterns can vary widely among zones. Noncollections zones and occupied collections zones tend to have high occupancy only at certain times, which may include events outside of normal operating hours. Some gallery occupancies are as high as 1 m^2 per person, but unoccupied storage zones may have 100 m^2 per person or less. Systems should be designed to handle maximum loads as well as the more common part loads. Many engineers design to 2 m^2 per person because part of the room is never occupied. In zones where spaces are extensively used for receptions, openings, and other high-traffic activities, even higher density assumptions may be justified. Continual and close